

Review

Using artificial neural networks to ask ‘why’ questions of minds and brains

Nancy Kanwisher,^{1,2} Meenakshi Khosla,^{1,2} and Katharina Dobs^{3,4,*}

Neuroscientists have long characterized the properties and functions of the nervous system, and are increasingly succeeding in answering how brains perform the tasks they do. But the question ‘why’ brains work the way they do is asked less often. The new ability to optimize artificial neural networks (ANNs) for performance on human-like tasks now enables us to approach these ‘why’ questions by asking when the properties of networks optimized for a given task mirror the behavioral and neural characteristics of humans performing the same task. Here we highlight the recent success of this strategy in explaining why the visual and auditory systems work the way they do, at both behavioral and neural levels.

Optimization as an answer to why questions about the living world

At the dawn of the 19th century the naturalist Alexander von Humboldt explored the Americas, describing in rich detail the characteristics of each plant and animal species he encountered. Thirty years later, his travelogue inspired the young Charles Darwin [1] to undertake his famous voyage on the Beagle, leading to a theory of why each species had the particular characteristics it did. Here we show how ANNs are helping to usher in a similar transformation in cognitive science and neuroscience, from a focus on describing phenomena of the mind and brain and their underlying mechanisms, to a deeply theoretical enterprise of asking (and sometimes even answering) why they work the way they do.

ANNs are simulated networks of neuron-like units that are optimized by extensive training on a particular task through gradual adjustment of the connection strengths between units (Figure 1). These networks thus enable us to test the hypothesis that a particular mental or neural phenomenon observed in humans results from optimization for a specific task, by asking first whether that phenomenon arises spontaneously in a network trained on that task, and then, crucially, whether it does not arise when the network is optimized for other tasks. Thus, much as evolutionary theory explains the shape of a beak or length of a neck as not simply arbitrary species characteristics, but solutions to specific biological problems optimized by natural selection, we can explain specific characteristics of mind and brain as optimized solutions for specific computational problems faced by organisms. In the case of minds and brains, though, the optimization can occur either through evolution or through learning during development, or (more often) a complex combination of the two. Both forms of optimization offer possible answers to why minds and brains work the particular ways they do.

The idea that the particular problems the brain must solve strongly influence the computations it conducts is not new. For instance, David Marr noted long ago that ‘the nature of the computations that underlie perception depends more upon the computational problems that have to be solved than on the particular hardware in which their solutions are implemented’ [2]. This idea is reflected in the concept of an ideal observer that performs optimally on a perceptual task given the available information [3], providing explanations for many observed visual phenomena

Highlights

Understanding the mind and brain requires determining not only how they work, but why they work the way they do.

We argue that artificial neural networks (ANNs) provide a new method for addressing ‘why’ questions about the brain.

If an ANN optimized for a given task spontaneously produces a particular phenomenon previously observed in humans, but optimization for other tasks does not, that suggests that the phenomenon may result from optimization of the brain for that same task.

We review phenomena in vision and audition (e.g., specific illusions), and of cortical organization (e.g., specializations for face recognition) that arise spontaneously in ANNs optimized for specific tasks, providing possible explanations for why these phenomena occur in humans.

Among the next goals for research in this area would be to discover the underlying principles that explain why each optimized ANN produces the particular human-like phenomenon it generates (the ‘why of the why’).

¹Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA

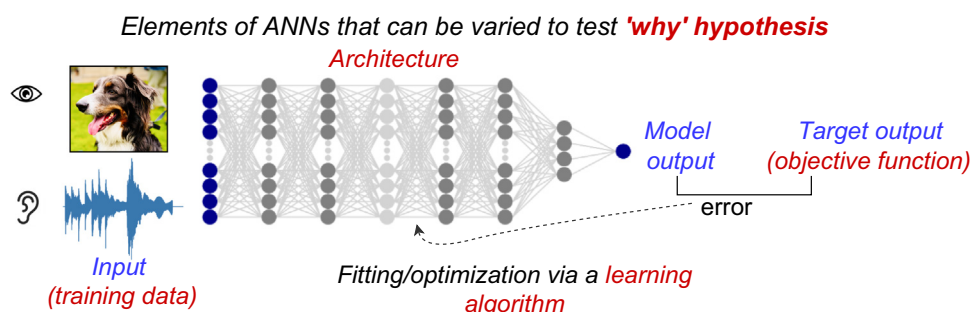
²McGovern Institute for Brain Research, Massachusetts Institute of Technology, Cambridge, MA, USA

³Department of Psychology, Justus Liebig University Giessen, Giessen, Germany

⁴Center for Mind, Brain and Behavior (CMBB), University of Marburg and Justus Liebig University, Giessen, Germany

*Correspondence: katharina.dobs@psychol.uni-giessen.de (K. Dobs).





Trends in Neurosciences

Figure 1. Dimensions of artificial neural networks (ANNs) that can be varied. Modern ANN models differ in four primary dimensions that can be manipulated independently. Answering why humans exhibit a particular behavioral or neural phenomenon entails determining which of these factors (or their combination) cause the ANN to exhibit that phenomenon. (i) The objective function is the task the ANN is optimized for. We focus primarily on the ability of task constraints to provide normative explanations for phenomena observed in humans. However, such findings are always within specifications of the other three dimensions that may also play an important role. (ii) Training data can be systematically varied to understand the role of input statistics in shaping a neural or perceptual phenomenon. For example, input statistics that mimic infant experience can inform what is learnable (and what properties result spontaneously) from sensory data alone without domain-specific inductive biases [73]. Varying degrees of realism [21] or ecological validity of category distributions [74] in training data can reveal how different perceptual/neural phenomena might be adapted to the constraints of the naturalistic, real-world environment. (iii) Architecture variations include the number and size of layers in the network, whether the network is purely feedforward or contains skip connections or recurrence [75,76], cell-type variability [77], and wiring costs [39]. Such variations can, for example, explain why the function of the retina differs between species [57]. (iv) Learning algorithms – the methods by which ANNs learn their tasks – can be pitted against each other in their ability to produce rich brain-like representations. More brain-like learning constraints include the use of unsupervised proxies as labels for training [55,56], biologically plausible plasticity rules for weight updates [78–80], or limited supervised training to better mimic the experience-dependent learning in primates [81]. Note that the learning approach and the objective function can be interdependent, as in the case of unsupervised learning. Photograph courtesy of Nancy Kanwisher.

in real brains. However, the ideal observer approach has proved less tractable for higher-level perceptual and cognitive processes. Here ANNs can help by discovering optimized (if not optimal) solutions to complex real-world computational problems [4]. When ANNs optimized for human-like tasks produce human-like phenomena, provides a possible explanation of why brains exhibit those phenomena, as well as an illustration of the very hardware independence Marr proposed.

One critical test of a causal explanation, the answer to a ‘why?’ question, is an intervention that removes the putative cause and asks if the effect still occurs [5]. A classic problem with evolutionary explanations of current species characteristics is that scientists cannot perform interventions that alter past objective functions and/or environmental constraints and rerun evolution (except in organisms with a very short generation time, like bacteria [6]). ANNs offer a solution to this problem for the case of optimization of minds and brains [7].

In this review we illustrate this general approach with examples of ANNs optimized for particular tasks that spontaneously produce known properties of minds and brains, thus explaining those properties as the possible results of optimizations for those tasks. We consider a broad definition of ANNs where an ANN is any network composed of simple computation units that loosely mimic real neurons, and where the connections weights between units are optimized according to an objective. The objective is again broadly construed, for example, the objective could be defined with reference to explicit supervision signals such as human-defined labels (as in supervised learning) or proxy tasks (as in self-supervised learning), or the objective could be to approximate the data distribution and capture the underlying structure of the data (as in generative modeling or

unsupervised learning). Although we focus on the role of task optimization, we note that ANNs also differ in other respects, including architecture, training data, and learning algorithms (Figure 1). These factors may interact with task constraints or may independently explain why or how a particular observed characteristic can arise.

Deep convolutional neural networks (CNN) as models for visual object recognition

In the field of computational vision, hand-engineered models of visual object recognition have been proposed for decades, but their performance lagged behind human abilities [8–10]. Then, in 2012, a deep CNN trained end-to-end on millions of labeled images burst on the scene with a performance close to that of humans on classification of real-world images [11]. CNNs thus offered the first plausible and image-computable models of how object recognition might work in the brain. Further, comparisons between such CNNs and primates showed a remarkable (though imperfect) match in their fine-grained behavioral performance [12] and in their internal activations [13–15]. Importantly, these networks were not trained to model primate object recognition, but only to classify images, making their fit to brains non-obvious and important. The (partial) fit tells us that these CNNs capture something about how vision works in the brain. It further suggests that merely optimizing for the same task can lead to similar solutions in brains and machines, despite the radical differences in their hardware and learning rules, supporting Marr's conjecture that the brain's visual algorithms are fundamentally shaped by the problems they are optimized to solve. This match between CNNs and brains thus also tells us something about why primate vision works the way it does: this is simply what an optimized solution to visual classification looks like! The success of CNN models of vision has inspired similar efforts in other domains, such as auditory perception [16,17] and language. Transformer-based large language models optimized for predicting the next word fit behavioral and neural data in humans, and the better the model performs on next word prediction the more closely it matches human data, suggesting that prediction may be part of what the human language system has been optimized for [18,19]. We next illustrate the power of ANNs to answer specific 'why' questions in cognitive science and neuroscience, with recent examples, starting with psychophysics.

Answering why questions about behavior

For over 150 years, psychophysicists have labored to characterize in detail the behavioral characteristics of human perceptual performance, documenting perceptual illusions and measuring precisely how visual acuity declines with stimulus eccentricity, how pitch discrimination is affected by the particular harmonics present in a tone, and how face recognition is affected by stimulus inversion. Testing whether ANNs optimized for certain tasks show similar phenomena (Figure 2) enables us to ask why humans exhibit these particular properties.

Audition

A pioneering study illustrating this strategy asked why human pitch perception exhibits the many well-established psychophysical characteristics it does [20]. To answer this question, the authors trained ANNs to estimate the fundamental frequency (F0), which is the perceptual correlate of pitch, from natural sound stimuli. The key finding was that ANNs that performed best at the task showed many of the classical characteristics of human pitch perception, such as changes in perceived F0 when stimuli were bandpass-filtered to control which harmonics were audible. Importantly, though, if the networks were instead trained on sounds without background noise, or sounds with unnatural spectra, these human-like behavioral signatures did not emerge. Similarly, if the input representation was altered so that it differed from that present in the auditory nerve, the model performed less well at pitch discrimination than humans. These and other findings thus explain many characteristics of human pitch perception as the result of optimization for

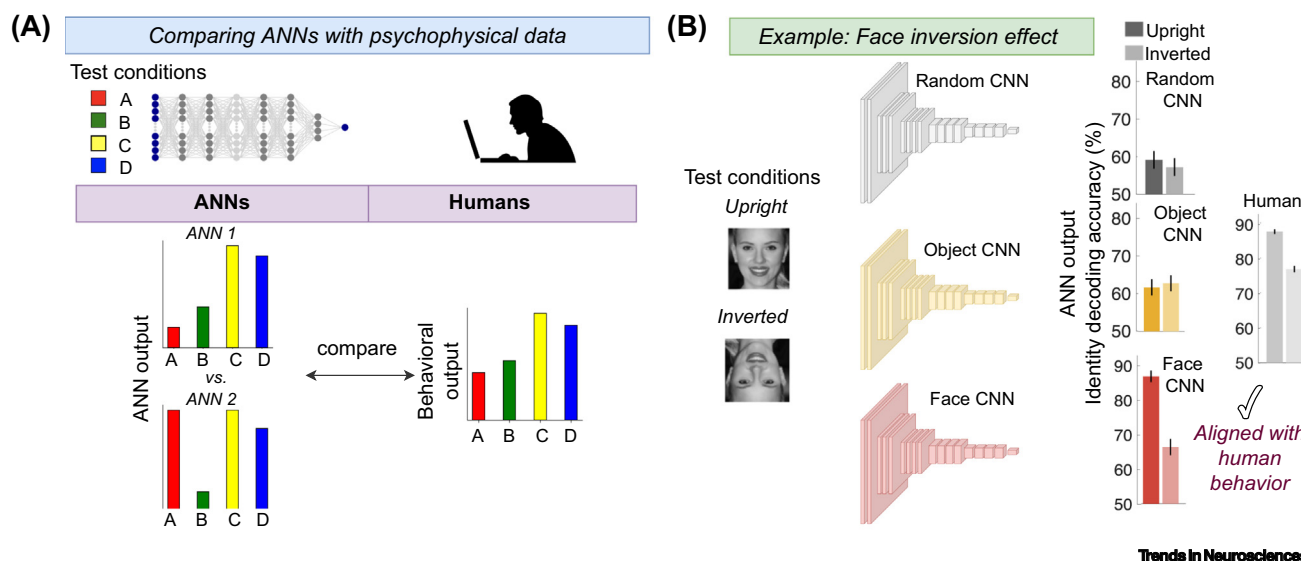


Figure 2. Comparing artificial neural networks (ANNs) with psychophysical data. (A) ANNs can be used to ask ‘why’ questions about human behavior by testing whether optimizing a network for a given task leads it to produce a known psychophysical phenomenon. (A) Behavioral phenomena can be measured in specific test conditions (e.g., A–D in red, green, yellow, and blue). By comparing the output of humans and ANNs to the same conditions, researchers can test which network properties lead to a particular phenomenon. (B) For example, ANNs can be used to ask why humans show a face-inversion effect [24], which is the lower accuracy for recognition of inverted faces compared to upright faces. When presenting ANNs that were either untrained (random convolutional neural network, CNN), trained on objects (object CNN), trained on faces (face CNN), with the same conditions and stimuli, only the face-trained CNN showed lower accuracy for inverted faces. This finding suggests that the face inversion effect in humans results from an optimization for upright face recognition, not from a general optimization to categorize objects. Face image from <https://commons.wikimedia.org>.

the specific problem of extraction of the fundamental frequency of real-world sounds given human cochlear input.

Another study in the same genre [21] asked why human sound localization exhibits the many specific psychophysical characteristics it does, such as frequency-dependent use of interaural time and level differences, localization dominance of sound onsets, and limitations on the ability to localize multiple concurrent sources. To find out, the authors built an ANN simulating the outer ears and head/torso with impulse responses recorded from a physical model of head and ears, and simulating a cochlea with a set of human-like bandpass filters. They then trained this model end-to-end to localize sounds generated in a virtual world with realistic background noise and reverberation. After training, they tested the model on a wide range of classic psychophysical tasks, and found that the model duplicated many previously established phenomena of human auditory localization, including those mentioned above. These findings suggest that many psychophysical properties of human auditory localization reflect optimizations for the specific problem of sound localization in natural environments given the fixed properties of the peripheral human auditory system.

Vision

Classic visual psychophysical phenomena are also starting to be explained as the result of optimizations for particular tasks. For example, many visual illusions arise spontaneously in networks optimized for visual tasks (Figure 3), and which illusions the CNNs replicate depends on the task each CNN is optimized to solve. Other psychophysical phenomena found in CNNs trained on object recognition (but not CNNs with random weights) include set size effects in visual search [22], a hallmark of human visual search performance, and human-like mirror confusion and scene incongruence effects (i.e., improved performance when an axe is embedded in a forest compared to a supermarket) [23], suggesting that these phenomena result from an optimization for object

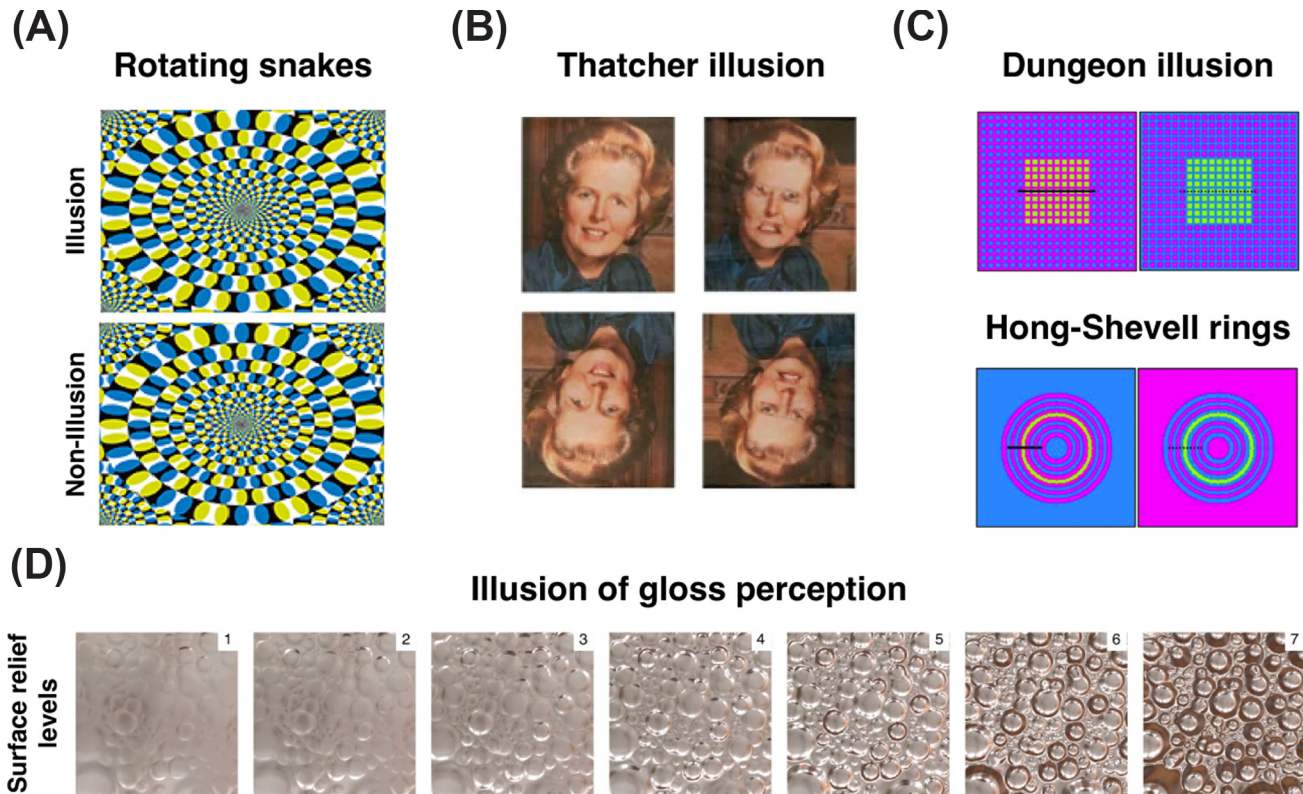


Figure 3. Why do humans experience visual illusions? Artificial neural networks (ANNs) are starting to reveal some of the specific network properties that lead to particular illusions. (A) The illusory motion apparent in the static 'rotating snake' image was mirrored in PredNet [82], a network trained to predict the next frame in thousands of video sequences [83], suggesting an account of this illusion in terms of predictive coding [84]. (B) The classic Thatcher illusion was present only weakly in convolutional neural networks (CNNs) trained on object recognition, but strongly in a CNN trained to recognize faces [23], suggesting that face-specific experience, and presumably optimization for face identification, is required to produce this effect. (C) Even very simple CNNs with only one hidden layer trained for basic low-level vision tasks – such as denoising, color constancy, and deblurring – replicate several classic illusions, such as the Dungeon illusion or the Hong–Shevell rings, in which the color of the central square or ring is biased toward the color of adjacent image regions [60]. (D) A generative ANN model trained to efficiently compress and spatially predict images of surfaces closely mimics illusions of gloss perception in humans, including the misperception that gloss increases with bumpiness (all seven images are from the identical material), suggesting a possible role of unsupervised learning objectives based on data compression in material perception in humans [85].

recognition in natural scenes. Importantly, however, object-trained CNNs did not show several properties of the human visual system related to 3D processing, occlusions or invariance to surfaces, and part-based processing [23], suggesting that these phenomena might result from other tasks beyond object classification. While these results support the role of specific tasks for properties of human object perception, a few phenomena of human object perception, such as relative size encoding, emerge in randomly initialized CNNs even before any training [23]. This finding emphasizes the role of network architecture (Figure 1) – in addition to the training task – and shows how random feedforward connections can already give rise to useful features.

Human face recognition exhibits a number of distinctive and well-documented behavioral 'signatures' such as overall high accuracy that drops significantly when faces are unfamiliar, or presented upside down, or originate from an ethnicity/race the observer is less familiar with. Why does face perception exhibit these properties? A recent study from our group used CNNs to test the hypothesis that these signatures result from an optimization for the task of fine-grained face discrimination [24]. We found that many of these face-processing signatures are

found in CNNs trained on face recognition, but not in CNNs trained on object recognition, suggesting that these phenomena may arise from optimization specifically for face recognition [23,25]. Alternatively, however, these properties could simply emerge from face-specific experience (without the need to discriminate the faces individually). To disentangle these two hypotheses, we manipulated the task and the amount of face-specific experience by training CNNs on face detection (categorizing all faces into one category) in addition to object recognition [24]. We found that when including the same amount of face experience, but without training the CNNs to perform fine-grained face discrimination, the classic human signatures of face perception were absent or weak, implicating optimization for face discrimination, rather than simply face experience, in these phenomena (Figure 2B).

Taken together, these findings demonstrate how varying the experience and task in CNNs enables us to go beyond documenting and reporting behavioral phenomena of the visual system, to asking why it exhibits these phenomena in the first place. Of course, simply saying that phenomenon X results from optimization for task Y begs the further question of why optimization for Y produces X, the 'why of the why', which can be further pursued in a variety of ways (Box 1).

Answering why questions about brains

Deep neural network models can inform 'why' questions not only about human behavior but also about the organization and function of the brain. Comparing ANNs optimized for certain tasks to minds and brains enables us to ask why the brain exhibits particular properties (Figure 4). We start with the earliest stage of the visual system and follow it downstream.

The organization of early stages of visual processing

Why do primates have much lower spatial resolution in the visual periphery than in the center of vision? This fact is usually explained in terms of the high metabolic and wiring cost of photoreceptors and their connections. But one study tested a different hypothesis by training a neural network on a visual search task entailing saccade-like translations of the input image over a receptor lattice [26]. The position and spatial resolution of each receptor was optimized over

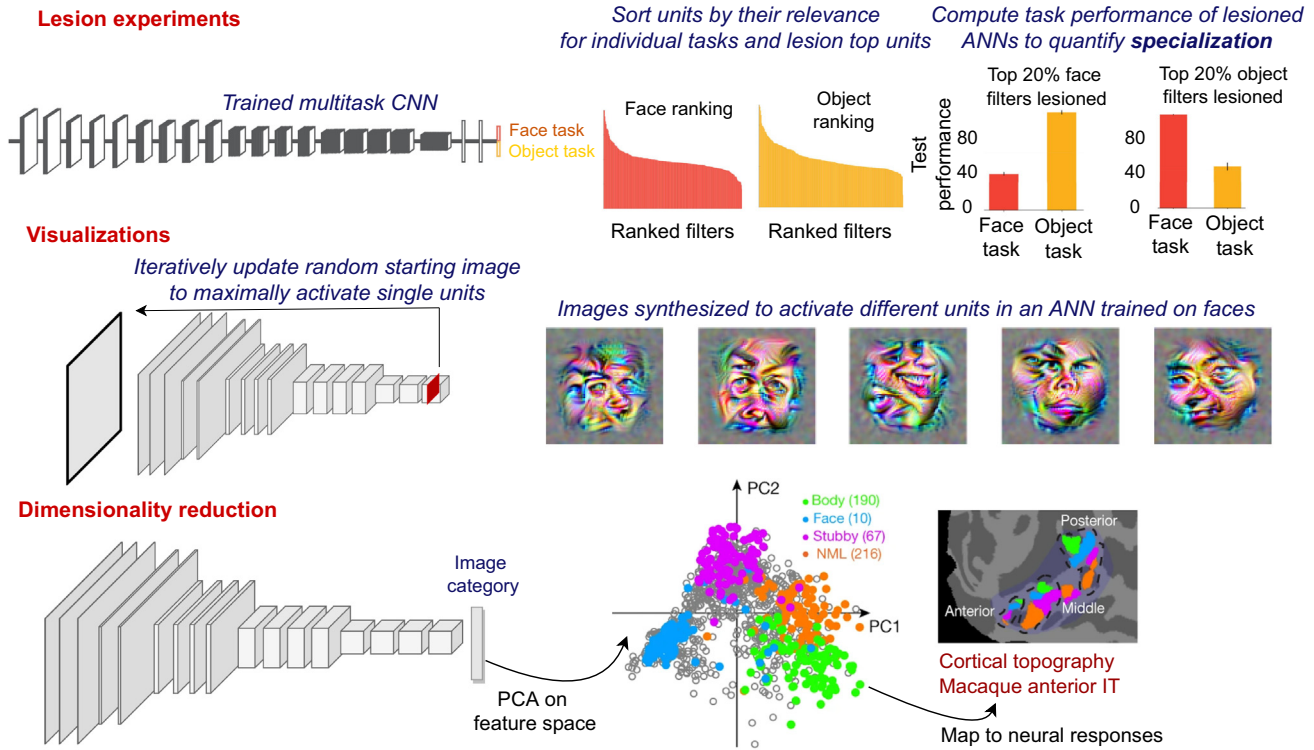
Box 1. The why of the why

Comparing brains to ANNs can reveal the design constraints that may have shaped brains, answering 'why' questions about observed neural or behavioral phenomena. However, simply invoking specific objective functions and constraints as explanatory primitives begs another question.

What specific aspects of a particular task are critical for an optimized ANN to reproduce a neural/behavioral phenomenon of interest? Two examples illustrate initial strategies for delving deeper into the 'why of the why'.

Evidence that specialization for face and object recognition in the brain results from optimization for both tasks comes from the finding that a similar segregation arises spontaneously in ANNs jointly trained on both tasks [44]. But this result begs the question of why face-processing segregates in both brains and networks (the 'why of the why'). That is, what is it about the task of face recognition that requires separate machinery? One hypothesis is that functional segregation will arise only for categories in which the exemplars to be discriminated all share the same basic shape. Initial evidence for this hypothesis was found when car discrimination segregated from object classification in networks trained on both, but no segregation was found for discrimination of faces from two data sets with different low-level image properties. However, segregation of the highly heterogeneous visual category of food in both networks [44] and brains [72] suggests that this account is incomplete at best.

In answer to why human sound localization is dominated by the first part of a sound (the 'precedence effect'), a recent study found that ANNs trained to localize naturalistic sounds show the same effect [21]. But why should this effect arise in both networks and brains? To test the hypothesis that the precedence effect represents a solution to the problem caused by echoes, which can reflect off surfaces far from the original sounds source, the authors trained a new ANN on sound localization using sounds generated in an anechoic environment. Consistent with their hypothesis, no precedence was observed in this network, suggesting that the effect represents a solution to the ambiguity caused by echoes.



Trends in Neurosciences

Figure 4. Methods for comparing artificial neural networks (ANNs) to minds and brains. Lesion experiments (illustrated in the top row) test the causal role of specific network units in ANN task performance, analogous to lesion studies in humans. Single unit analyses (not illustrated) in networks and brains reveals the tuning properties of individual units, whether they are selective for meaningful perceptual attributes, forming disentangled representations [86], or exhibit mixed selectivity and distributed representations [87]. Analyses in this vein have revealed the spontaneous emergence of face-selective [88,89] and number-selective units [90,91] in ANNs. Visualizing (illustrated in middle row) stimuli that maximally activate convolutional neural network (CNN) units provides an intuitive understanding of their selectivity [92], allowing comparison to neurons. Dimensionality reduction techniques (illustrated in bottom row) like principal component analysis (PCA) reveal the low-dimensional structure of representations in ANNs and brains [93]. Comparing task performance of ANNs (not illustrated) varying in architecture can reveal which architecture provides a better foundation upon which to learn specific tasks. For example, the number of shared layers can be varied in branched ANNs trained on two tasks (dual-task network) to test whether those tasks are best performed together or separately [16]. Representational distance comparisons (not illustrated): the distance between the vector of response across a set of ANN units to two different stimuli can be used as a measure of the representational dissimilarity of those two stimuli in a network, which can then be compared to representational dissimilarity in minds and brains (e.g., in the Thatcher effect where scrambled and unscrambled faces are more dissimilar when upright than inverted) [23].

training, and produced a higher-resolution ‘fovea’ at the center and a lower-resolution periphery. Importantly, when the model was allowed to make non-biological image transformations like zooming, the fovea-like organization did not arise, suggesting that the organization of the primate fovea may have evolved in part to enable efficient sampling of visual information over saccades. Another study showed that CNNs trained on object recognition with a human blur profile at the input stage outperformed networks trained on input images with steeper or shallower blur profiles, or full resolution images [27]. Both findings suggest that a blurred visual periphery may be a feature, not a bug, reflecting evolutionary optimization for object recognition and/or efficient sampling of visual information across eye movements.

Moving along the visual pathway, another recent study asked why V1 has the functional characteristics it does, by building these properties into the front of a CNN (VOneNet) and training it on Imagenet [28]. The resulting network was more robust to adversarial attacks and common image corruptions than state-of-the-art networks. Further, each of the properties of V1 built into the

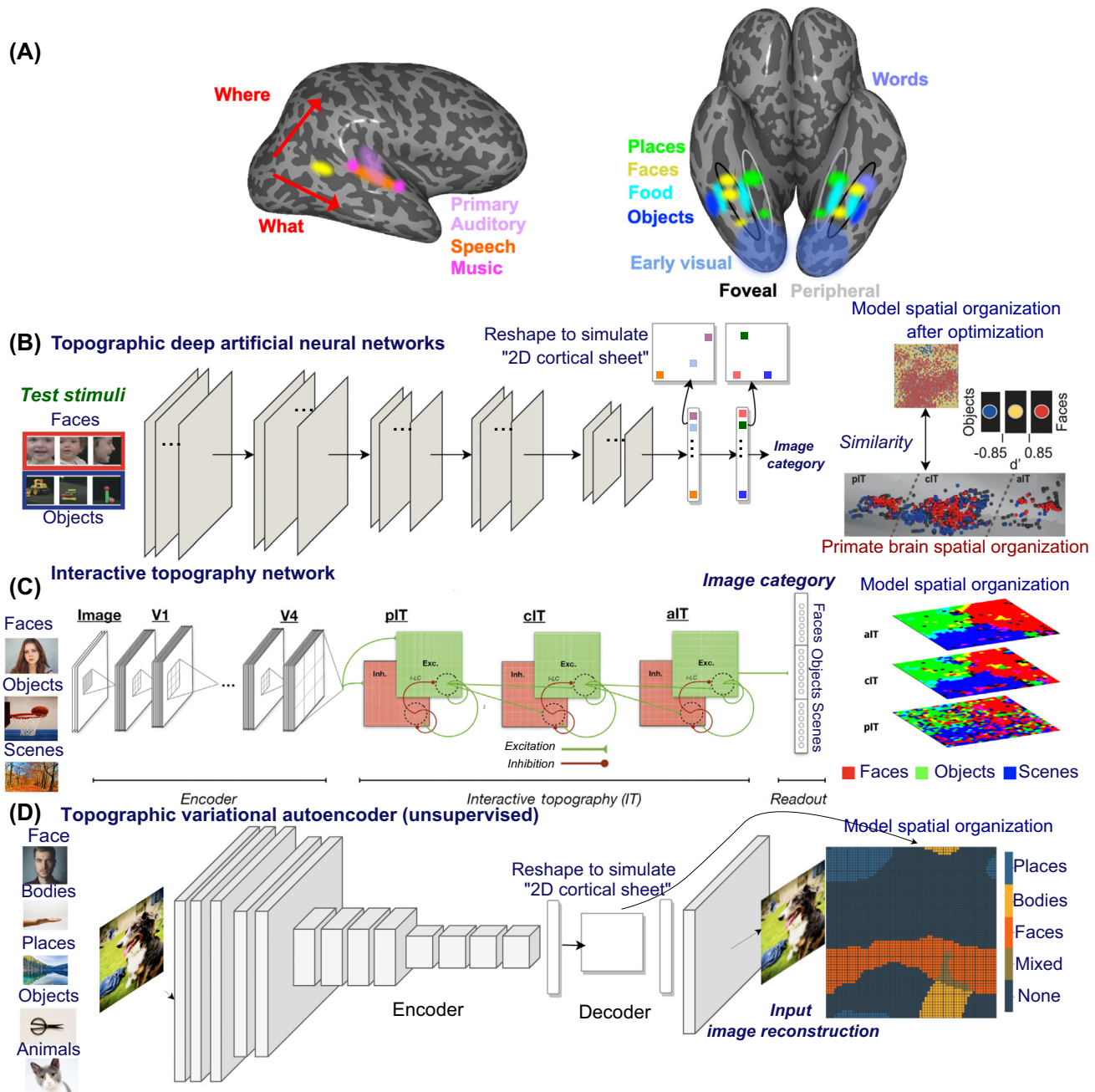
model contributed to the network's robust performance, as removal of any one of them decreased robustness. These results suggest that many properties of V1 reflect evolutionary optimizations for robust image classification. Another study tested whether processing of color and luminance would spontaneously segregate in CNNs trained on object recognition, as they do in early visual cortex [29]. The authors trained multiple instances of Alexnet and found a high degree of segregation of chromatic and achromatic information across CNN instances. Moreover, the degree of segregation in a network was correlated with its performance. This finding suggests that the segregation of color and luminance processing in the human visual cortex may also result from optimization for real-world object recognition.

Another study attempted to understand the systematic spatial organization of early visual cortex using computational models built on self-organizing principles [30]. Specifically, by transforming the visual input space into a tuned 2D map, with each unit tuned to some aspect of the visual space and nearby units having similar tuning, a repeating map topography emerged, similar to the primate visual system. This suggests a role for biologically plausible self-organizing principles (which in turn reflect approximate solutions to wiring cost minimization) in shaping the organization of the early visual cortex (Figure 5A).

Higher-level stages of visual cortex

From retinotopic cortex, visual processing diverges into a ventral object recognition (or 'what') stream and a dorsal ('where') stream processing object location and visually guided action. One of the first studies of the genre highlighted in this review asked why the visual cortex is organized into these two streams [31]. To find out, the authors trained two versions of a simple three-layer connectionist network: one in which the nodes in the hidden layer were split between those connected only to the shape output nodes and others connected only to the location output nodes, and another version in which all hidden units were connected to all output nodes. The authors found that the split networks outperformed the unsplit network, but only when more hidden units were allocated to the (more difficult) 'what' task. A related connectionist study used a modular architecture in which the different modules compete to learn the task, resulting a partitioning of the task into multiple functionally distinct subtasks, with a distinct module allocated to each [32]. This problem was later revisited with modern deep neural networks by training CNNs for two visual tasks simultaneously [33]. Specifically, the authors manipulated the relatedness of the two tasks to test the hypothesis that the need to perform two unrelated tasks (like the 'what' and 'where' task during object processing) results in the emergence of segregated processing streams dedicated to each task. In the CNN trained on related tasks, the majority of units contributed to both tasks, whereas in the CNN optimized for unrelated tasks, units often contributed disproportionately to a specific task, and the degree of specialization increased with progressive layers. Taken together, these and other findings [34] suggest that segregation of function in the visual system results from the optimization for multiple tasks with different computational goals (Figure 5A).

Other recent studies have asked why the ventral visual pathway is organized the way it is, with small regions selective for the specific categories of faces [e.g., occipital face area (OFA), fusiform face area (FFA)] [35], places [parahippocampal place area (PPA)] [36], bodies [extrastriate body area (EBA)] [37], and words [visual word form area (VWFA)] [38] embedded in larger cortical regions exhibiting gradients of weak preferences for mid-level features (e.g., [39–41]) (Figure 5B–D). One such study [42] trained a CNN on object and scene categorization and found that the network organized itself into units with a central image bias and units more selective for the background of images. This finding suggests that the dissociation between fovea-biased and periphery-biased regions of the ventral pathway [43], and the functional properties of each, may reflect an optimization for the classification of images from different visual categories (Figure 5A).



Trends in Neurosciences

Figure 5. Aspects of cortical organization that have been informed by artificial neural network (ANN) models. (A) ANN models trained on basic perceptual tasks provide computational explanations for multiple aspects of observed cortical organization. ANNs mirror the functional organization of the early visual cortex [30] and hierarchical organization of auditory cortex [16,17], show layer-wise correspondences between neural responses in primary versus higher-level regions of visual cortex [13–15], and brain-like functional differences between fovea-biased and periphery-biased cortex in the ventral pathway [42]. Functional dissociations emerge in ANNs for faces versus objects and food versus objects [44], for the processing of visual words [45], for ‘what’ versus ‘where’ pathways in vision [31,33,34], and speech versus music in auditory cortex [16]. Several ANN models with connectivity constraints and topographic representations have further been used to account for the systematic spatial organization of the high-level visual cortex. These models primarily differ in what connectivity constraints are assumed and how the ANN representational space is mapped to the two-dimensional topographic space. (B) Localized face selectivity emerged in a topographic ANN model trained for

(Figure legend continued at the bottom of the next page.)

In another recent study, we asked why the visual cortex exhibits such a high degree of specialization, and why it does so for some categories but apparently not for others [44]. Using network lesioning methods (Figure 4), we found that a CNN jointly trained on both face and object recognition spontaneously segregated itself into separate systems for faces and objects, suggesting that human brains show this organization as a result of joint optimization for both tasks (Figure 5A and Box 1). In a related line of work, ANNs were harnessed to account for the emergence of visual word selectivity in the ventral visual cortex [45]. The authors simulated the late acquisition of reading abilities in humans by training an ANN model in two phases, first on general image recognition, and subsequently on both image and word recognition. A subset of single units exhibited strong word-selectivity following training, thereby capturing known properties of the VWFA (Figure 5A). This model provides a further demonstration of the computational utility of functional segregation, as well as supporting the cortical-recycling theory of development by demonstrating how a network optimized for generic object recognition may be repurposed to additionally perform visual word recognition by co-opting a small subset of units for this task.

Within the domain of face perception, a classic view [46] holds that while face identity is processed by the ventral pathway (e.g., including OFA and FFA), facial expressions are processed in a lateral temporal pathway including the posterior superior temporal sulcus (STS). However, this view has been challenged by reports that facial expressions can also be decoded from ventral areas, and identity information can be decoded from lateral regions [47,48]. Recently, CNNs were used to inform this debate. If segregation of facial identity and expression processing is required from a computational point of view, then CNNs trained to recognize identities should outperform CNNs trained to recognize facial expressions on face identification and vice versa. Interestingly though, expression-selective units spontaneously emerge in CNNs trained for facial identity (and vice versa) [49,50], and these expression-selective units show human-like characteristics [51]. These results suggest that face identity and expression are processed interdependently, and that functional segregation of these two processes is not necessarily expected on computational grounds. Moreover, this finding shows how CNNs can be used not only to ask why some processes are functionally segregated, but also to explain why other processes are integrated.

Functional organization of auditory cortex

Can the functional organization of auditory cortex also be understood as resulting from task optimization? Recent studies have trained CNNs on auditory tasks and found correspondences between the trained models and human auditory cortex, with earlier stages predicting primary auditory regions and deep stages predicting nonprimary regions [16,17,52,53]. Further, models trained on multiple auditory tasks had the best overall predictivity for neural responses [17]. Training CNNs that branched at different layers on speech recognition and musical genre classification further revealed that the networks that performed best shared early processing stages across tasks, but engaged separate pathways for speech and music at later stages [16]. This branched network matched human task performance, exhibited a similar pattern of errors, and predicted voxel responses in human auditory cortex, despite being optimized only for task performance. This work suggests that the functional organization of human auditory cortex reflects optimizations for the human auditory tasks including speech and music recognition (Figure 5A).

supervised image categorization under approximate wiring constraints that cause nearby units to exhibit correlated activity [39]. (C) A recurrent ANN model trained on object, scene, and face categorization with an explicit wiring cost function developed brain-like cortical organization with selective patches for faces, scenes, and objects [40]. (D) Localized category-selective organization for faces, places, and bodies can emerge even with unsupervised learning instantiated in the form of a topographic variational autoencoder [41]. Dog photograph courtesy of Nancy Kanwisher.

Unsupervised training to ask ‘why’ questions about development

An important new direction in the use of ANNs to inform human behavioral and neural organization is the recent advent of unsupervised models trained not on labeled data but on suitable unsupervised proxies for labels that are obtained via simple image manipulations. The representations learned via these ‘semi-supervised’ learning models now compete with supervised models in their object recognition behavior [54]. Recent papers showed that ANNs trained with deep contrastive unsupervised methods can predict neural responses to images in the monkey [55] and human [56] ventral visual pathway as well as supervised models, and also exhibit more human-like error patterns [55] than supervised models. This finding suggests that the biologically unrealistic form of label-based feedback received by supervised models is not necessary to achieve human-like neural or perceptual phenomena, filling in the explanatory gap created by supervised models which rely on millions of semantic labels and are thus implausible as models of biological learning. All these unsupervised models sample different views (augmentations) of the same image and are trained to maximize agreement between their representations; unlike supervised models, the labels for semi-supervised ANN models (i.e., the views) are accessible in humans, for example, through retinal distortions or saccades [56]. While unsupervised learning algorithms are thus clearly more biologically plausible than their supervised counterparts, both techniques still align in an overarching goal, for example by providing proxy labels to achieve object recognition. Importantly, these models can help us get closer to answering ‘why’ questions from the lens of postnatal development during which labels are rarely provided.

Strengthening evidence for optimization arguments

The research strategy described here entails inferring that when an ANN optimized for a given task spontaneously produces behavioral or neural characteristics previously described in humans, but optimization for other tasks does not produce the same characteristics, that suggests that these human characteristics reflect optimization for that task. It is important to note that no claims about mechanistic similarity between ANNs and brains are made in this framework since it is the optimization, and not the precise mechanism, that is driving the explanations. Different models could have the same input–output function but vary in their mechanistic plausibility; yet if they all exhibit the neural/behavioral phenomenon of interest, then this suggests an important link between optimization for the function and the emergent phenomenon.

As in any scientific domain, we can never definitively prove that an optimization hypothesis is true, and future evidence could always overturn it. However, the strength of the evidence that phenomenon X resulted from task optimization Y will increase with (i) the breadth of networks varying in hyperparameters but sharing the same task optimization Y that all produce phenomenon X, and crucially, (ii) the range of networks optimized for different tasks that do not produce X. We can further attempt to find the ‘minimal’ sufficient condition for phenomenon X to emerge in ANNs, that is, the smallest possible difference between ANNs that produce a cognitive/neural phenomenon and ANNs that do not. Moreover, it is important to determine whether the effect in the network is of similar magnitude to that observed in humans, such that its emergence in networks provides a compelling explanation of the human phenomenon. Finally, we can increase the rigor of the overall scientific program outlined here by reducing experimenter degrees of freedom with preregistrations of our specific optimization hypotheses and the ANNs we will use to test them.

Further, there may be a few cases in which we can experimentally test a proposed answer to a ‘why’ hypothesis. When the answer refers to optimization over development, it is sometimes possible to create in animals, or find in humans, conditions in which the relevant experience differs. In these cases, we can test whether developmental optimization that hinges on the availability of

particular experience can explain the emergence of a given phenomenon. For the case of optimization over evolution one can sometimes test hypothesized answers to ‘why’ questions that appeal to optimization by comparing across species with different niches [57].

Limitations of optimization arguments

While the research strategy described here is potentially powerful, it is subject to several important caveats. For one, if the characteristic in question emerges only for certain sets of network hyperparameters, the argument loses force, showing merely that the characteristic can emerge from the objective function in question, rather than that it is either likely or sure to. It is therefore essential that researchers do not cherry-pick among the large space of possible hyperparameters if they are trying to use ANNs to answer ‘why’ questions. Better yet, researchers should kick the tires on network hyperparameters (e.g., using different random initializations [58]) to test the robustness of the emergence of the particular phenomenon in question. Second, even when the characteristic in question does not emerge in control models (e.g., an ANN with random initialization or a different training objective), a thorough optimization of all relevant parameters for the control models is essential to make general statements about the failure of control conditions to yield the phenomenon in question [59].

Further, because multiple objective functions can in principle share a common solution, a model that replicates observed neural responses is not guaranteed to have the same objective function as the brain. By contrast, in some cases where multiple task optimizations produce the same characteristic [60], researchers can attempt to understand what those tasks have in common such that they produce that characteristic or what other emergent phenomena are consistently shared across these diverse ANNs [61], and researchers can then test those hypotheses by training networks with new objective functions. Indeed, this approach will ultimately enable us to explore the ‘why of the why’ by not just testing which task optimizations lead to a given solution, but what exactly it is about that task or training set that leads to that solution (Box 1). The converse challenge is that a given objective function may have multiple solutions (i.e., multiple local minima), so there is no guarantee that even the correct objective function will lead to a match to the brain. One proposal for addressing this problem is the ‘contravariance principle’ according to which the space of possible solutions is smaller for more complex tasks, suggesting that the approach advocated here may be most effective when applied to higher-level perceptual and cognitive processes [62]. In any case, the upshot of these critiques is that while ANNs are powerful tools that are opening up new avenues for addressing long-standing why questions in cognitive science and neuroscience (see Outstanding questions), they cannot do the job on their own, and we still need to think hard about the computational principles underlying the emergence of phenomena from network optimization.

Concluding remarks

We have argued here that much as evolution offers a framework for explaining why organisms have the characteristics they do, ANNs give us a method for asking why the human brain has the characteristics it does. Importantly, though, the solutions found in both evolution and ANNs are optimized but not optimal: evolution depends on prior conditions and a dynamically changing environment, and optimization in ANNs is always tested within a particular set of hyperparameters. Nonetheless, these two frameworks enable scientists to move beyond the mere collection of facts, and the exploration of underlying mechanisms, to approach some of the deepest theoretical questions about why organisms and minds work the way they do. And whereas evolutionary theories are sometimes criticized as ‘just so’ stories because of the difficulty of testing those theoretical accounts, with ANNs researchers can test their explanations by altering the objective function, learning rules, training data, or architecture. We have described here some exciting first steps

Outstanding questions

Most of the research sketched here focuses on phenomena that emerge from optimization for task performance. To what extent does incorporating detailed biological constraints like wiring costs, energy efficiency, and cell-type variability improve the ability to reproduce the emergence of various phenomena?

Much of the past work in the area has asked whether and when human-like functional organization, or human-like representational geometry, emerges from ANNs. To what extent will ANNs spontaneously produce phenomena at the level of individual neurons based on the current abstractions of biological neurons in ANNs? If unsuccessful in that regard, what additional principles will be needed to capture the response properties of individual neurons? For example, would more neuron-like properties such as spiking be required?

What inductive biases – such as priors on architectures and training frameworks – need to be built into an ANN for human-like phenomena to emerge from training? For example, given typical human visual input, will human-like behavioral signatures of face recognition automatically result, or will this happen only given a mechanism to direct attention to faces?

Recent progress in unsupervised learning has achieved performance and neural predictivity comparable to supervised methods. To what extent will such generic learning strategies lead to human-like behavioral and neural phenomena? Will unsupervised learning dynamics spontaneously recapitulate developmental trajectories characteristic of humans?

How much of the functional organization of the brain can be explained by the principle of optimization for multiple human tasks? How far can end-to-end deep learning lead in explaining high-level cognitive phenomena? Would these approaches be able to fully recapitulate high-level cognitive phenomena, or might other approaches – such as embodied artificial intelligence models that instead learn through interactions from their environment – be needed?

where this strategy has been successfully applied to a wide array of behavioral phenomena such as auditory pitch perception and sound localization, visual illusions, and higher-level visual processing of objects and faces. It has also illuminated multiple aspects of the functional organization of visual and auditory processing in the cortex. Opportunities abound for extending this work into new domains, including language [18,19,63], navigation [64–67], motor control [68,69], and higher-level cognition [70,71]. There is grandeur in this lens on cognition, which enables us to query the characteristics of our very own minds and brains with the quintessentially human question: why?

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Declaration of interests

The authors declare no competing interests in relation to this work.

References

- Hiroshi, S. (2017) A. Wulf: the invention of Nature. *Geogr. Rev. Jpn. Ser. A* 90, 625–626
- Marr, D. (1982) *Vision*, Freeman
- Geisler, W.S. (2011) Contributions of ideal observer theory to vision research. *Vis. Res.* 51, 771–781
- Kell, A.J. and McDermott, J.H. (2019) Deep neural network models of sensory systems: windows onto the role of task constraints. *Curr. Opin. Neurobiol.* 55, 121–132
- Pearl, J. and Mackenzie, D. (2018) *The Book of Why: The New Science of Cause and Effect*, Basic Books
- Barrick, J.E. and Lenski, R.E. (2013) Genome dynamics during experimental evolution. *Nat. Rev. Genet.* 14, 827–839
- Richards, B.A. *et al.* (2019) A deep learning framework for neuroscience. *Nat. Neurosci.* 22, 1761–1770
- Viola, P. and Jones, M. (2004) Robust real-time face detection. *Int. J. Comput. Vis.* 57, 137–154
- Riesenhuber, M. and Poggio, T. (1999) Hierarchical models of object recognition in cortex. *Nat. Neurosci.* 2, 1019–1025
- Fukushima, K. (1980) Neocognitron: a self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biol. Cybern.* 36, 193–202
- Krizhevsky, A. *et al.* (2012) ImageNet classification with deep convolutional neural networks. *Adv. Neural Inf. Proces. Syst.* 25, 1097–1105
- Rajalingham, R. *et al.* (2018) Large-scale, high-resolution comparison of the core visual object recognition behavior of humans, monkeys, and state-of-the-art deep artificial neural networks. *J. Neurosci.* 38, 7255–7269
- Yamins, D.L.K. *et al.* (2014) Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proc. Natl. Acad. Sci. U. S. A.* 111, 8619–8624
- Khaligh-Razavi, S.-M. and Kriegeskorte, N. (2014) Deep supervised, but not unsupervised, models may explain IT cortical representation. *PLoS Comp. Biol.* 10, e1003915–29
- Güçlü, U. and Gerven, M.A.J. van (2015) Deep neural networks reveal a gradient in the complexity of neural representations across the ventral stream. *J. Neurosci.* 35, 10005–10014
- Kell, A.J.E. *et al.* (2018) A task-optimized neural network replicates human auditory behavior, predicts brain responses, and reveals a cortical processing hierarchy. *Neuron* 98, 630–644.e16
- Tuckute, G. *et al.* (2022) Many but not all deep neural network audio models capture brain responses and exhibit hierarchical region correspondence. *bioRxiv* Published online November 05, 2022. <https://doi.org/10.1101/2022.09.06.506680>
- Schrimpf, M. *et al.* (2021) The neural architecture of language: Integrative modeling converges on predictive processing. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2105646118
- Goldstein, A. *et al.* (2022) Shared computational principles for language processing in humans and deep language models. *Nat. Neurosci.* 25, 369–380
- Saddler, M.R. *et al.* (2021) Deep neural network models reveal interplay of peripheral coding and stimulus statistics in pitch perception. *Nat. Commun.* 12, 7278
- Franci, A. and McDermott, J.H. (2022) Deep neural network models of sound localization reveal how perception is adapted to real-world environments. *Nat. Hum. Behav.* 6, 111–133
- Nicholson, D.A. and Prinz, A.A. (2021) Deep neural network models of object recognition exhibit human-like limitations when performing visual search tasks. *bioRxiv* Published online January 12, 2021. <https://doi.org/10.1101/2020.10.26.354258>
- Jacob, G. *et al.* (2021) Qualitative similarities and differences in visual object representations between brains and deep networks. *Nat. Commun.* 12, 1872
- Dobs, K. *et al.* (2022) Using deep convolutional neural networks to test why human face recognition works the way it does. *bioRxiv* Published online November 24, 2022. <https://doi.org/10.1101/2022.11.23.517478>
- Blauch, N.M. *et al.* (2020) Computational insights into human perceptual expertise for familiar and unfamiliar face recognition. *Cognition* 208, 104341
- Cheung, B. *et al.* (2017) Emergence of foveal image sampling from learning to attend in visual scenes. *arXiv* Published online October 21, 2017. <https://doi.org/10.48550/arXiv.1611.09430>
- Pramod, R.T. *et al.* (2022) Human peripheral blur is optimal for object recognition. *Vis. Res.* 200, 108083
- Dapello, J. *et al.* (2020) Simulating a primary visual cortex at the front of CNNs improves robustness to image perturbations. *bioRxiv* Published online October 22, 2020. doi.org/10.1101/2020.06.16.154542
- Flachot, A. and Gegenfurtner, K.R. (2018) Processing of chromatic information in a deep convolutional neural network. *J. Opt. Soc. Am. A* 35, B334–B346
- Konkle, T. (2021) Emergent organization of multiple visuotopic maps without a feature hierarchy. *bioRxiv* Published online January 06, 2021. doi.org/10.1101/2021.01.05.425426
- Rueckl, J.G. *et al.* (1989) Why are what and where processed by separate cortical visual systems? A computational investigation. *J. Cogn. Neurosci.* 1, 171–186
- Jacobs, R.A. *et al.* (1991) Task decomposition through competition in a modular connectionist architecture: The what and where vision tasks. *Cogn. Sci.* 15, 219–250
- Scholte, H.S. *et al.* (2018) Visual pathways from the perspective of cost functions and multi-task deep neural networks. *Cortex* 98, 249–261

34. Bakhtiari, S. *et al.* (2021) The functional specialization of visual cortex emerges from training parallel pathways with self-supervised predictive learning. *Adv. Neural Inf. Proces. Syst.* 34, 25164–25178
35. Kanwisher, N. *et al.* (1997) The fusiform face area: a module in human extrastriate cortex specialized for face perception. *J. Neurosci.* 17, 4302–4311
36. Epstein, R. and Kanwisher, N. (1998) A cortical representation of the local visual environment. *Nature* 392, 598–601
37. Downing, P.E. *et al.* (2001) A cortical area selective for visual processing of the human body. *Science* 293, 2470–2473
38. Cohen, L. *et al.* (2000) The visual word form area: spatial and temporal characterization of an initial stage of reading in normal subjects and posterior split-brain patients. *Brain* 123, 291–307
39. Lee, H. *et al.* (2020) Topographic deep artificial neural networks reproduce the hallmarks of the primate inferior temporal cortex face processing network. *bioRxiv* Published online July 10, 2020. <https://doi.org/10.1101/2020.07.09.185116>
40. Blauch, N.M. *et al.* (2022) A connectivity-constrained computational account of topographic organization in primate high-level visual cortex. *Proc. Natl. Acad. Sci. U. S. A.* 119, e2112566119
41. Keller, T.A. *et al.* (2021) Modeling category-selective cortical regions with topographic variational autoencoders. *arXiv* Published online December 19, 2021. <https://doi.org/10.48550/arXiv.2110.13911>
42. Mohsenzadeh, Y. *et al.* (2020) Emergence of visual center-periphery spatial organization in deep convolutional neural networks. *Sci. Rep.* 10, 1–8
43. Levy, I. *et al.* (2001) Center-periphery organization of human object areas. *Nat. Neurosci.* 4, 533–539
44. Dobs, K. *et al.* (2022) Brain-like functional specialization emerges spontaneously in deep neural networks. *Sci. Adv.* 8, eab18913
45. Hannagan, T. *et al.* (2021) Emergence of a compositional neural code for written words: recycling of a convolutional neural network for reading. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2104779118
46. Haxby, J.V. *et al.* (2000) The distributed human neural system for face perception. *Trends Cogn. Sci.* 4, 223–233
47. Klemann, D. *et al.* (2018) Cortical responses to dynamic emotional facial expressions generalize across stimuli, and are sensitive to task-relevance, in adults with and without autism. *Cortex* 103, 24–43
48. Dobs, K. *et al.* (2018) Task-dependent enhancement of facial expression and identity representations in human cortex. *NeuroImage* 172, 689–702
49. O'Neill, K.C. *et al.* (2019) Recognition of identity and expressions as integrated processes. *PsyArXiv* Published online May 21, 2019. <http://doi.org/10.31234/osf.io/9c2e5>
50. Colón, Y.I. *et al.* (2021) Facial expression is retained in deep networks trained for face identification. *J. Vis.* 21, 4
51. Zhou, L. *et al.* (2022) Emerged human-like facial expression representation in a deep convolutional neural network. *Sci. Adv.* 8, eabj4383
52. Vaidya, A.R. *et al.* (2022) Self-supervised models of audio effectively explain human cortical responses to speech. *arXiv* 2205.14252
53. Millet, J. *et al.* (2022) Toward a realistic model of speech processing in the brain with self-supervised learning. *arXiv* 2206.01685
54. Goyal, P. *et al.* (2019) Scaling and benchmarking self-supervised visual representation learning. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6390–6399
55. Zhuang, C. *et al.* (2021) Unsupervised neural network models of the ventral visual stream. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2014196118
56. Konkle, T. and Alvarez, G.A. (2022) A self-supervised domain-general learning framework for human ventral stream representation. *Nat. Commun.* 13, 491
57. Lindsey, J. *et al.* (2019) A unified theory of early visual representations from retina to cortex through anatomically constrained deep CNNs. *arXiv* Published online January 3, 2019. <http://doi.org/10.48550/arXiv.1901.00945>
58. Mehrer, J. *et al.* (2020) Individual differences among deep neural network models. *Nat. Commun.* 11, 5725
59. Funke, C.M. *et al.* (2021) Five points to check when comparing visual perception in humans and machines. *J. Vis.* 21, 16
60. Gomez-Villa, A. *et al.* (2019) Convolutional neural networks can be deceived by visual illusions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12301–12309
61. Maheswaranathan, N. *et al.* (2019) Universality and individuality in neural dynamics across large populations of recurrent networks. *Adv. Neural Inf. Proces. Syst.* 32, 15629–15641
62. Cao, R. and Yamins, D. (2021) Explanatory models in neuroscience: Part 2 – constraint-based intelligibility. *arXiv* 2104.01489
63. Caucheteux, C. and King, J.-R. (2022) Brains and algorithms partially converge in natural language processing. *Commun. Biol.* 5, 134
64. Banino, A. *et al.* (2018) Vector-based navigation using grid-like representations in artificial agents. *Nature* 557, 429–433
65. Sorscher, B. *et al.* (2022) A unified theory for the computational and mechanistic origins of grid cells. *Neuron* 111, 121–137.e13
66. Schaeffer, R. *et al.* (2022) No free lunch from deep learning in neuroscience: a case study through models of the entorhinal-hippocampal circuit. *bioRxiv* Published online August 7, 2022. <https://doi.org/10.1101/2022.08.07.503109>
67. Sorscher, B. *et al.* (2022) When and why grid cells appear or not in trained path integrators. *bioRxiv* Published online November 15, 2022. <https://doi.org/10.1101/2022.11.14.516537>
68. Kalidindi, H.T. *et al.* (2021) Rotational dynamics in motor cortex are consistent with a feedback controller. *eLife* 10, e67256
69. Sussillo, D. *et al.* (2015) A neural network that finds a naturalistic solution for the production of muscle activity. *Nat. Neurosci.* 18, 1025–1033
70. Yang, G.R. *et al.* (2019) Task representations in neural networks trained to perform many cognitive tasks. *Nat. Neurosci.* 2, 297–306
71. Musslick, S. and Cohen, J.D. (2021) Rationalizing constraints on the capacity for cognitive control. *Trends Cogn. Sci.* 25, 757–775
72. Khosla, M. *et al.* (2022) A highly selective response to food in human visual cortex revealed by hypothesis-free voxel decomposition. *Curr. Biol.* 32, 4159–4171.e9
73. Orhan, A.E. (2021) How much 'human-like' visual experience do current self-supervised learning algorithms need to achieve human-level object recognition? *arXiv* Published online May 24, 2022. <http://doi.org/10.48550/arXiv.2109.11523>
74. Mehrer, J. *et al.* (2021) An ecologically motivated image dataset for deep learning yields better models of human vision. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2011417118
75. Kietzmann, T.C. *et al.* (2019) Recurrence is required to capture the representational dynamics of the human visual system. *Proc. Natl. Acad. Sci. U. S. A.* 116, 21854–21863
76. Kar, K. *et al.* (2019) Evidence that recurrent circuits are critical to the ventral stream's execution of core object recognition behavior. *Nat. Neurosci.* 22, 974–983
77. Perez-Nieves, N. *et al.* (2021) Neural heterogeneity promotes robust learning. *Nat. Commun.* 12, 5791
78. Lillicrap, T.P. *et al.* (2016) Random synaptic feedback weights support error backpropagation for deep learning. *Nat. Commun.* 7, 13276
79. Scellier, B. and Bengio, Y. (2017) Equilibrium propagation: bridging the gap between energy-based models and backpropagation. *Front. Comput. Neurosci.* 11, 24
80. Bartunov, S. *et al.* (2018) Assessing the scalability of biologically-motivated deep learning algorithms and architectures. *Adv. Neural Inf. Proces. Syst.* 31, 9390–9400
81. Geiger, F. *et al.* (2020) Wiring up vision: minimizing supervised synaptic updates needed to produce a primate ventral stream. *bioRxiv* Published online June 08, 2020. <https://doi.org/10.1101/2020.06.08.140111>
82. Watanabe, E. *et al.* (2018) Illusory motion reproduced by deep neural networks trained for prediction. *Front. Psychol.* 9, 345
83. Lotter, W. *et al.* (2017) Deep predictive coding networks for video prediction and unsupervised learning. *arXiv* Published online March 1, 2017. <https://doi.org/10.48550/arXiv.1605.08104>
84. Rao, R.P. and Ballard, D.H. (1999) Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nat. Neurosci.* 2, 79–87
85. Storrs, K.R. *et al.* (2021) Unsupervised learning predicts human perception and misperception of gloss. *Nat. Hum. Behav.* 5, 1402–1417
86. Higgins, I. *et al.* (2021) Unsupervised deep learning identifies semantic disentanglement in single inferotemporal face patch neurons. *Nat. Commun.* 12, 6456

87. Fong, R. and Vedaldi, A. (2018) Net2Vec: quantifying and explaining how concepts are encoded by filters in deep neural networks. *arXiv* Published online March 29, 2018. <https://doi.org/10.48550/arXiv.1801.03454>
88. Baek, S. *et al.* (2021) Face detection in untrained deep neural networks. *Nat. Commun.* 12, 7328
89. Xu, S. *et al.* (2021) The face module emerged in a deep convolutional neural network selectively deprived of face experience. *Front. Comput. Neurosci.* 15, 626259
90. Kim, G. *et al.* (2021) Visual number sense in untrained deep neural networks. *Sci. Adv.* 7, eabd6127
91. Nasr, K. *et al.* (2019) Number detectors spontaneously emerge in a deep neural network designed for visual object recognition. *Sci. Adv.* 5, eaav7903
92. Zeiler, M.D. and Fergus, R. (2014) Visualizing and understanding convolutional networks. *Eur. Conf. Comput. Vis.* 8689, 818–833
93. Bao, P. *et al.* (2020) A map of object space in primate inferotemporal cortex. *Nature* 583, 103–108